

6)

Cannot change h without also changing x

Decreasing the mass of the particles, for example to 0.06, causes them to all drop down in a straight line

Increasing the mass of the particles, for example to 5.00, causes them to “burst” and fly quickly in all directions from the corner

Smaller ρ (100.0) causes particles to burst and scatter

Larger ρ (10000.0) initializes particles already bouncing around slowly

Small k (300.0) makes particles drop down slightly to bottom of frame then stop

Large k (30000.0) makes columns of particles expand away from each other slightly then stop

Larger μ (1.1) causes columns of particles to stick together like a chain

Smaller μ (0.01) causes particles to drop down slightly and stop

Smaller interval for g ([0.0, -1.8]) makes columns of particles expand away from each other slightly then stop

Larger interval for g ([0.0, -100.8]) causes simulation to run much faster

Positive interval for g ([0.0, 9.8]) causes particles to float up slightly and stop

Small dt (2.0e-10) particles do not move

Large dt (2.0e-1) only a few particles visible moving rapidly mostly at the bottom of frame

Small STEPS (1000) particles move very slight amount downward and stop

Large STEPS (500000) similar to original

SAVE_EVERY (100) simulation very slow

SAVE_EVERY (1) simulation very slow and stops early

Large dx (5.8) only 2 particles present

Small dx (0.4) large number of particles, rapid related expansion and contraction

7)

When using cubic spline the particles “burst” and some form small clumps

8)

450 particles. The particles still scatter a bit but overall they behave more like the original simulation.

9)

The time complexity for spatial hashing is $O(n)$ because the time complexity for searching a hash map is $O(1)$. Therefore, naive search takes more time, which would be more noticeable for large numbers of particles.

10)

Euler-cromer updates the velocity values while Euler uses the previous values. The Euler method causes the magnitude of the eigenvalues to be greater than 1. Thus the algorithm is less stable because the oscillations are not bounded.